

2081

Bachelor of Science in Computer Science and Information Technology

Course Title: **Database Management System**

Code No: **CSC265** Semester: **4th Semester**

Full Marks: 60 Pass Marks: 24 Time: 3 hours

1. Consider the library database schema containing the tables given below where primary keys are underlined.

USER (UserID, User_name, Address) **USES** (UserID, BookID, Issued_date, Returned_date) **BOOK** (BookID, Book_name, Author, PublisherID) **PUBLISHER** (PublisherID, Publisher_name, Address)

Specify the following queries in both SQL and Relational Algebra on this database schema.

a. Retrieve names of all books issued to the user "Hari".

b. Retrieve the names of all books published by "ABC publication".

c. Count the number of books published by the author "Ram".

3. What are desirable properties of transactions? Explain conflict serializability with example.

Section B

Attempt any Eight questions [8×5=40]

4. What do you mean by referential integrity? Why do you need this integrity? Explain with example.

5. Explain ANSI/SPARC three-schema architecture with suitable diagram.

6. What are informal design guidelines for relational schemas?

Candidates are required to answer the questions in their own words as far as practicable.

Section A

- 1. What do you mean by entity type and entity set? Construct an ER diagram for an airline ticket booking system where customers buy airline tickets online. The system should provide discounts on the basis of number of tickets bought. It should also keep records of buyers on the basis of number of times buyer visits the system. It should also filter the unwanted visitors to manage traffic congestion.**
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- 7.** Classify database management systems based on the data model.
 - 8.** Define the terms data, database, DBMS, database system, and database catalog.
 - 9.** Explain relational algebra natural join (*) operation with example.
 - 10.** Explain deferred update approach in database recovery.
 - 11.** Explain transaction deadlock with example.
 - 12.** Define specialization. Explain disjoint constraint with example.